# Digital Systems

## Lecture 10

## Programming with Interrupts

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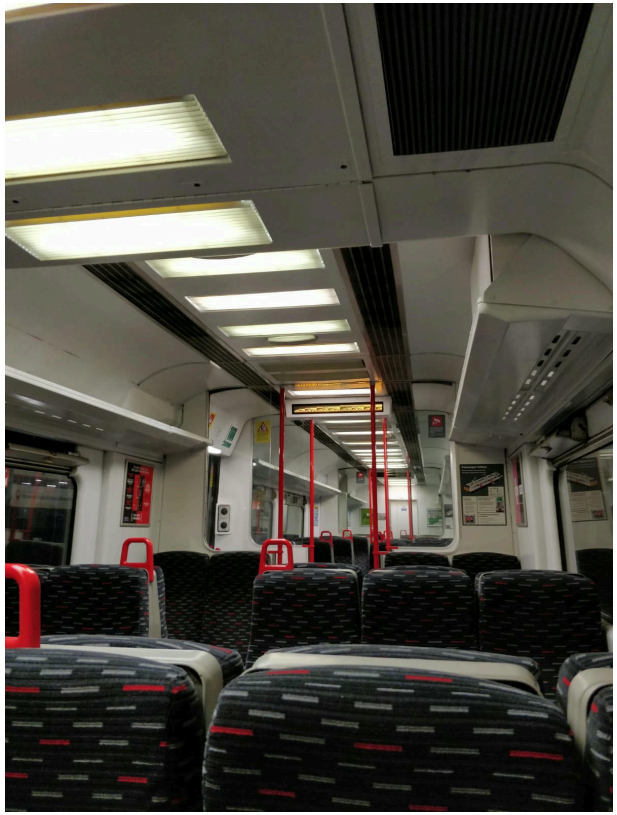
Hilary Term 2025

Department of
COMPUTER
SCIENCE

Following the course of past years by Mike Spivey

# More Chocolate



What is going on here? Chocolate for the best hypothesis.

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**Hardware should operate at the same time as CPU is executing useful instructions.**

There was a *tiny* bit of parallelism: The last character to be sent!

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- to have a minimal waiting time to respond to an event,
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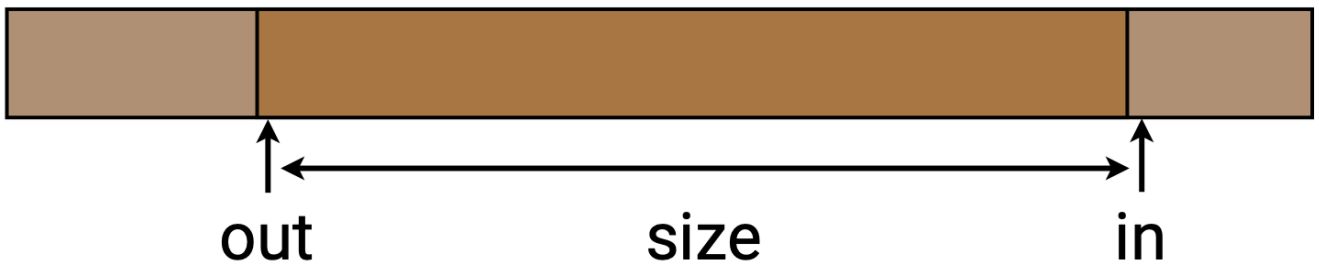
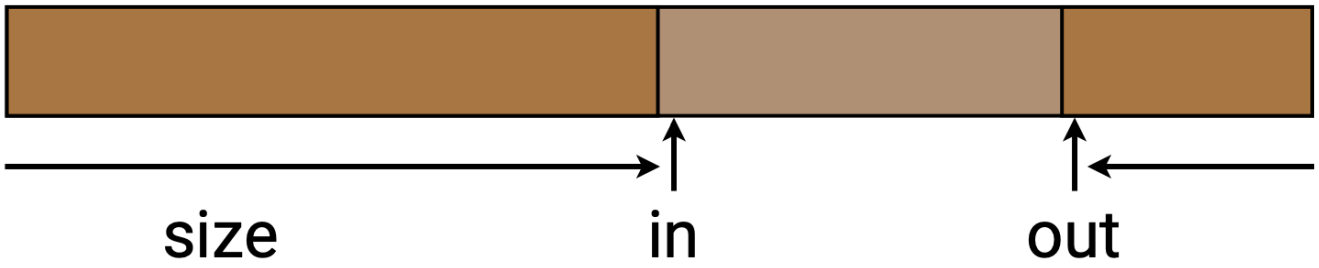
```
#define NBUF 64
static int bufcnt = 0;
static int bufin = 0;
static int bufout = 0;
static char txbuf[NBUF];
```

# Circular Buffer




- Add characters at `bufin`.

- Send characters from `bufout`.

# One Possible Solution

```
void serial_putc(char ch) {
  while (bufcnt == NBUF) { pause(); }   // Buffer is full...
  txbuf[bufin] = ch; bufcnt++;
  bufin = (bufin+1) % NBUF;
}
```

```
int prime(int n) {
    int k = 2;

    while (k * k <= n) {
        if (n % k == 0) return 0;
        poll_uart();
        k++;
    }

    return 1;
}
```

```
void poll_uart(void) {
    if (UART_TXDRDY) {
        // send another char
    }
}
```

# What do we think of this solution?

We want:

- 
- 
- 
- 

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- ❓ to have a minimal waiting time to respond to an event,
- ❌ software that is modular, in the sense that it doesn't need to be adapted when other parts of the program change which events they deal with.

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Hardware *interrupts* the normal execution of instructions in response to an event, and starts executing a different subroutine.

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- Variables that are shared between interrupt and main code, must be marked `volatile` (can change any time, outside direct code execution).

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Hardware *interrupts* the normal execution of instructions in response to an event, and starts executing a different subroutine.

- When `UART_TXDRDY` becomes true, execute `uart_handler()`.
- Variables that are shared between interrupt and main code, must be marked `volatile` (can change any time, outside direct code execution).

```
#define NBUF 64
static volatile int bufcnt = 0;
static int bufin = 0;
static int bufout = 0;
static volatile char txbuf[NBUF];
static volatile int txidle;
```

# Better Solution: Interrupts

```
void uart_handler(void) {
  if (UART_TXDRDY) {
    UART_TXDRDY = 0;
    if (bufcnt == 0) {
      txidle = 1;
    } else {
      UART_TXD = txbuf[bufout];
      bufcnt--;
      bufout = (bufout+1) % NBUF;
    }
  }
}
```

# Better Solution: Interrupts

```
void serial_putc(char ch) {
  while (bufcnt == NBUF) pause();//volatile ensures fresh read
  intr_disable();
  if (txidle) {
    UART_TXD = ch;
    txidle = 0;
  } else {
    txbuf[bufin] = ch; bufcnt++;
    bufin = (bufin+1) % NBUF;
  }
  intr_enable();
}
```

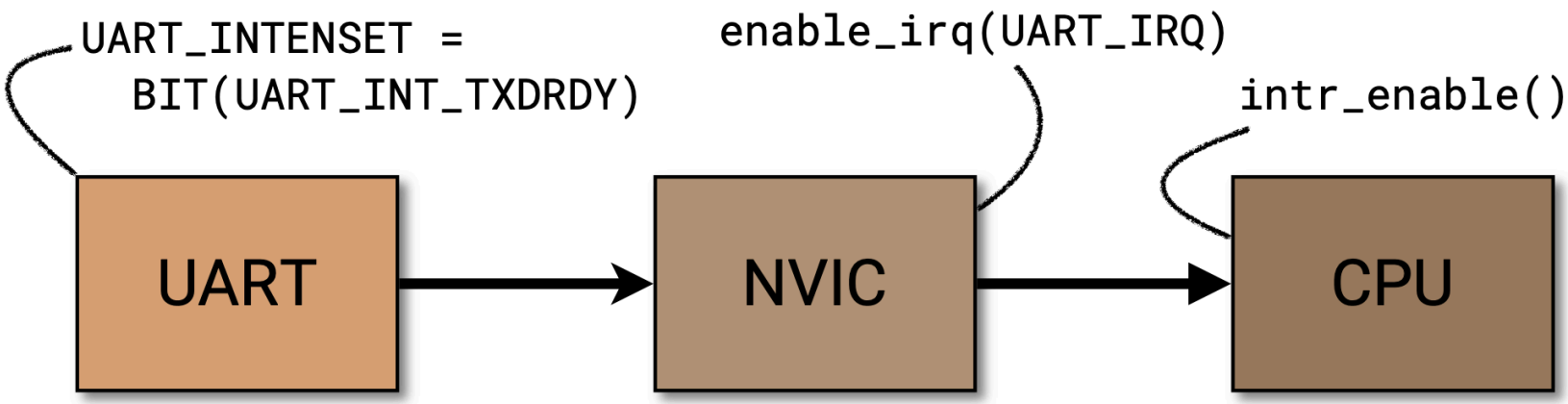
# Why Disable Interrupts?

- What happens if interrupt occurs before `bufcnt++;`?
- What happens if interrupt occurs

```
ldr r0, =bufcnt
ldr r1, [r0]
                        @ <== here?
add r1, r1, #1
str r1, [r0]
```

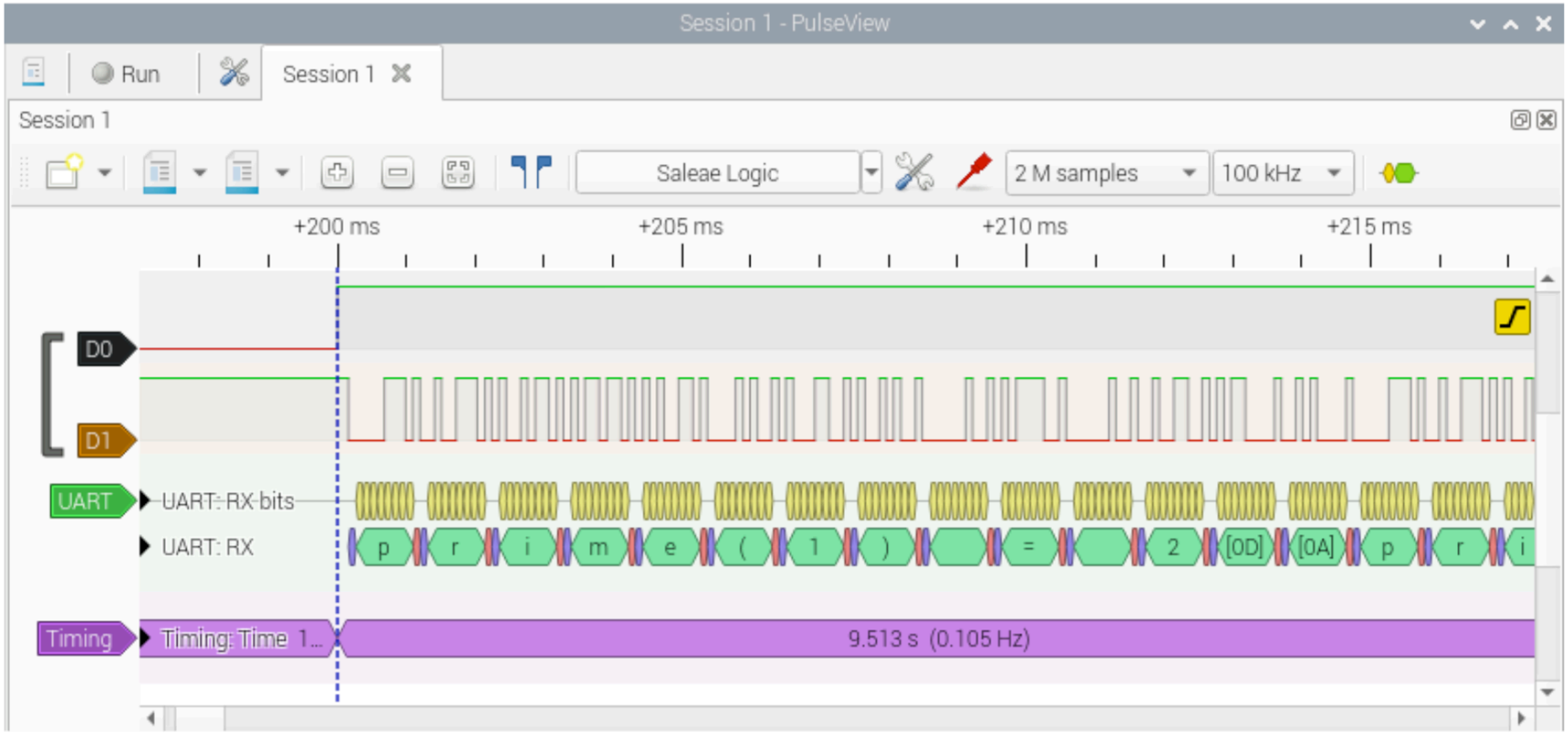
# Setting Up Interrupts

```
void serial_init(void) {
  ...
  UART_INTENSET = BIT(UART_INT_TXDRDY);
  enable_irq(UART_IRQ);
  txidle = 1;
}
```



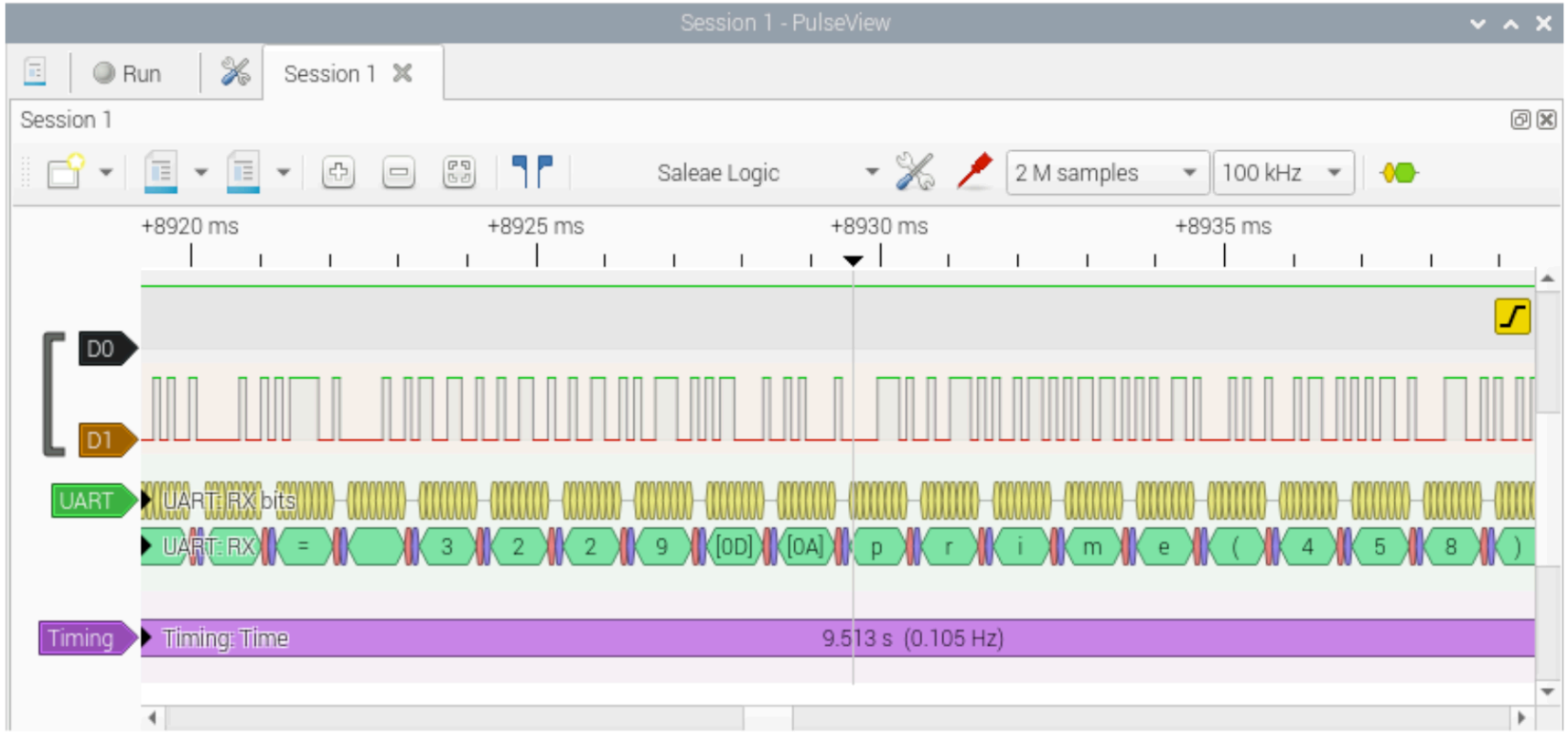
# Timings: Early on in Primes Program

No (noticeable) delays between sending characters.



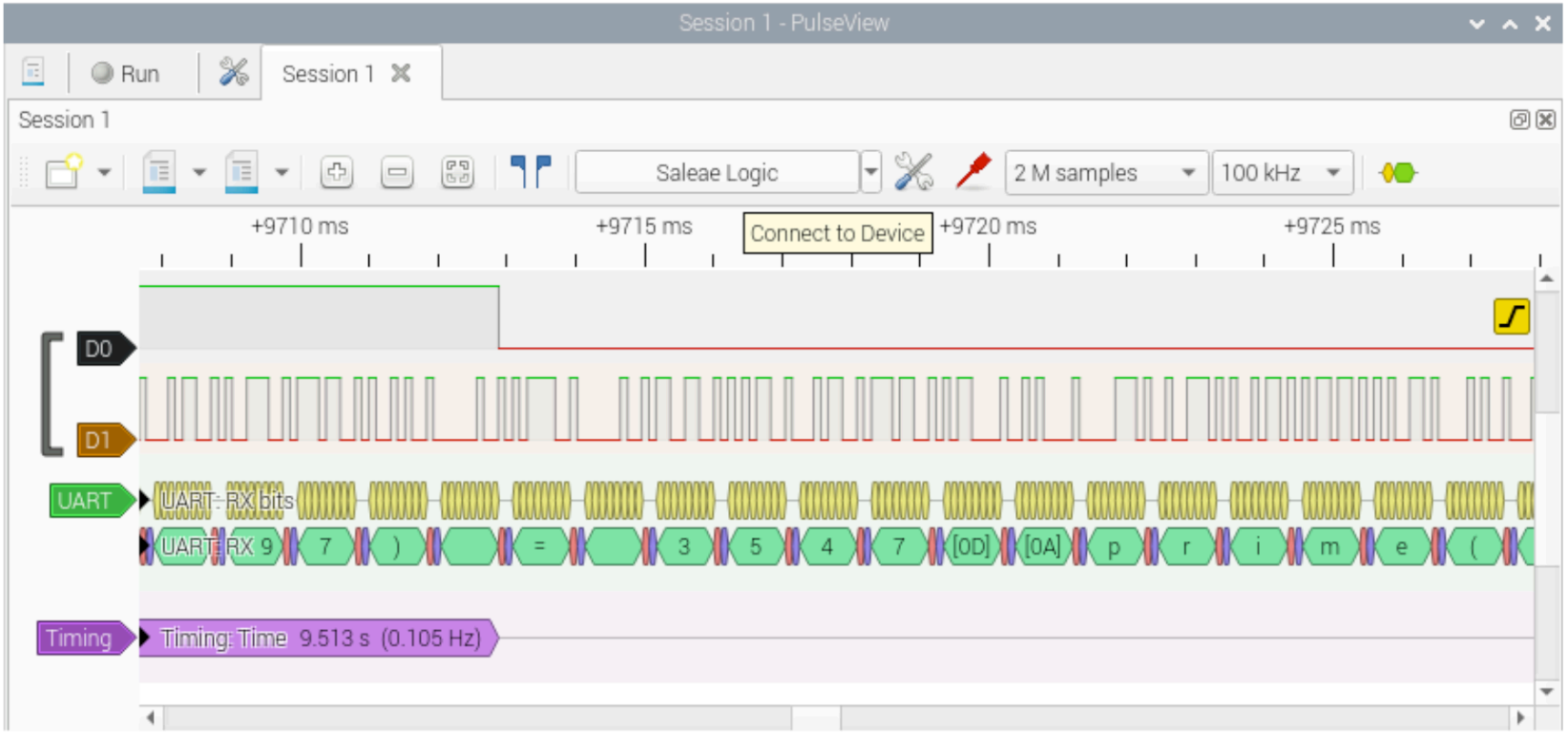
# Timings: Later on in Primes Program

No (noticeable) delays between sending characters.



# Timings: End of Primes Program

Sending characters continues after primes calculation is done!



**Crucial detail omitted!**

## Summary

- How to implement a queue with a circular buffer.
- Why we need interrupts.
- What hardware provides for interrupts.
- How to use interrupts in software.